

How-To: ragent Configuration (`ragent.json`)

ragent is configured through a single JSON file called `ragent.json`. This document is the complete reference for every section of that file: what it controls, the schema, the defaults, and worked examples.

If you only want to get running quickly, read the **Quick Start** section in the project root `README.md` and the **Tutorial** in `docs/howtos/tutorial.md`. This document is the deep reference.

Table of Contents

1. [Overview](#)
2. [File Discovery and Loading Order](#)
3. [Merge Semantics](#)
4. [CLI and Environment Overrides](#)
5. [Config Error Reporting](#)
6. [Backup and Restore](#)
7. [Section Reference](#)
 - 7.1 `username`
 - 7.2 `defaultAgent`
 - 7.3 `provider`
 - 7.4 `agent`
 - 7.5 `permission`
 - 7.6 `command`
 - 7.7 `mcp`
 - 7.8 `instructions`
 - 7.9 `skill_dirs`
 - 7.10 `experimental`
 - 7.11 `hooks`
 - 7.12 `bash`
 - 7.13 `dirs`
 - 7.14 `tool_visibility`
 - 7.15 `code_index`
 - 7.16 `stream`
 - 7.17 `memory`
 - 7.18 `compaction`
 - 7.19 `gitlab`
 - 7.20 `hidden_tools`
 - 7.21 `yolo`
 - 7.22 `edit_log`
 - 7.23 `prices`
 - 7.24 `browser`
 - 7.25 `channels`
 - 7.26 `gmail`
 - 7.27 `telemetry`
 - 7.28 `agent_perf`
 - 7.29 `finance`

- 7.30 `tavily_api_key` / `langsearch_api_key` / `perplexity_api_key` / `exa_api_key` / `openalex_email`
- 7.31 `sdd`
- 7.32 `trigger`
- 7.33 `piegap`
- 7.34 `research`

8. [Full Example File](#)

9. [Common Recipes](#)

10. [Related Documents](#)

1. Overview

All ragent runtime behaviour is driven by a single layered JSON configuration:

- **Provider connections** (API keys, base URLs, thinking settings, model catalogues)
- **Agent defaults and overrides** (system prompts, max steps, skills)
- **Permissions** (allow / deny / ask rules, bash and directory allow/deny lists)
- **Tool visibility** (which tool families are advertised to the LLM)
- **Memory, compaction, and retrieval** settings
- **Telemetry / OpenTelemetry export**
- **External integrations** (GitLab, Telegram/Discord channels, Gmail, finance data providers)
- **Feature flags** (experimental, SDD, pie-gap, triggers, research)

The config is loaded once at startup by `ragent-config::Config::load`, merged across layers, and then held in memory for the lifetime of the process. Several subsystems (YOLO mode, edit logging, bash/directory lists) also read runtime atomic flags that are synced from the config at startup.

Design principles

Principle	Meaning
Layered merge	Global config provides defaults; project config overrides; env vars override both.
Explicit-over-default	Many fields track whether the user <i>explicitly</i> set them so a serialise round-trip does not lose a toggle.
Sensible defaults	Every section has <code>Default</code> implementations so an empty <code>{}</code> is valid.
No secrets in config by force	API keys come from env vars or <code>env:VAR_NAME</code> indirection; the config file can reference them but does not have to store them.
Atomic persistence	Writes use temp-file-then-rename so a crash never leaves a partial config.

2. File Discovery and Loading Order

`Config::load` resolves configuration in a strict precedence order. Each layer is merged on top of the previous one, with later layers overriding earlier ones for explicitly-set fields.

```
compiled defaults
-> global file:  <config_dir>/ragent/ragent.json
```

```
-> project file:  ./..ragent/ragent.json
-> env file:      $RAGENT_CONFIG (path to a JSON file)
-> env inline:    $RAGENT_CONFIG_CONTENT (raw JSON string)
```

Platform config directory

The global config directory is resolved by the `dirs` crate:

Platform	Path
Linux	<code>\$XDG_CONFIG_HOME/ragent/ragent.json</code> (default <code>~/.config/ragent/ragent.json</code>)
macOS	<code>~/Library/Application Support/ragent/ragent.json</code>
Windows	<code>%APPDATA%\ragent\ragent.json</code>

The helper `Config::global_config_dir()` returns `<config_dir>/ragent` and `Config::global_config_path()` returns the full `ragent.json` path.

Project config

The project config lives at `./..ragent/ragent.json` relative to the current working directory. This is the file created automatically on first run when neither a global nor project config exists.

JSONC note

The crate-level documentation mentions `ragent.jsonc` as a supported filename, and JSONC-style comments appear in documentation examples (`//` line comments). However, the parser uses `serde_json::from_str`, which is strict JSON. Inline `//` comments are **not** supported in the actual config file. Use a `.json` extension and plain JSON. The `jsonc` examples in code doc-comments are for illustration only.

Auto-creation

If no config file is found at either the global or project path, `Config::load` creates a `..ragent/` directory and writes a default `ragent.json` there so the user has a starting point.

3. Merge Semantics

`Config::merge(base, overlay)` combines two configs. Different sections use different merge strategies:

Section	Strategy
<code>username</code> , API keys, <code>gitlab</code> , <code>channels</code> , <code>gmail</code>	Overlay overrides base when <code>Some</code> .
<code>defaultAgent</code>	Overlay overrides when explicitly set or different from default.
<code>provider</code>	Deep merge per provider key: models, API, and thinking are merged so partial overlays do not discard lower-layer defaults.
<code>agent</code> , <code>command</code> , <code>mcp</code>	Per-key overlay replaces base entry.

Section	Strategy
permission, instructions, skill_dirs, hooks	Append (union).
bash.allowlist, bash.denylist	Union (deduplicated).
prices	Per-model last-wins (overlay entry replaces base entry for the same model id).
code_index, tool_visibility	Overlay wins only for explicitly-set fields. The specified flags are propagated so the value survives a reload.
compaction	Overlay replaces base entirely.
yolo, edit_log	OR semantics: if the overlay sets true , it wins.
sdd, piegap	OR semantics: a flag enabled in either layer stays enabled.
research.open_access_recovery	OR semantics. contact_email and oa_min_full_text_chars : overlay overrides when present.
telemetry	If overlay enables telemetry, the whole overlay otel block replaces the base. Otherwise maps (resource_attributes , metrics) are unioned.
experimental	open_telemetry and parallel_tool_calls : OR semantics. Other fields: overlay defaults.

The **config_paths** field records every file that contributed to the final merged config, in load order.

4. CLI and Environment Overrides

--config <PATH> CLI flag

```
ragent --config /path/to/my-config.json run "hello"
```

When **--config** is supplied, the file at that path is loaded directly and bypasses the normal discovery order. Parse errors for this path include the file path, line, column, the problematic source line, and a caret marker.

Environment variables

Variable	Purpose
RAGENT_CONFIG	Path to a JSON config file, merged on top of project config.
RAGENT_CONFIG_CONTENT	Raw JSON string, merged last (highest precedence). Parse errors show the inline content.
Provider API keys	See section 7.3 . Env vars always take precedence over config-file values.
TAVILY_API_KEY	Overrides tavily_api_key .

Variable	Purpose
<code>OPENALEX_EMAIL</code>	Overrides <code>openalex_email</code> .
<code>EXA_API_KEY</code>	Overrides <code>exa_api_key</code> .
<code>GITLAB_TOKEN</code> , <code>GITLAB_URL</code> , <code>GITLAB_USERNAME</code>	Override <code>gitlab</code> fields (highest priority).

Other CLI flags that affect config

Flag	Effect
<code>--model <provider/model></code>	Overrides the resolved model.
<code>--agent <name></code>	Overrides <code>defaultAgent</code> .
<code>--yes</code> / <code>--no-prompt</code>	Auto-approves all permissions (like YOLO but does not persist).

5. Config Error Reporting

When a config file fails to parse, ragent produces an actionable diagnostic:

```
Failed to parse config file '.ragent/ragent.json':
Error at line 12, column 8:
```

```
_____
Problematic line:
```

```
    "defaultAgent: "coder"
      ^
```

```
Parse error: expected `:` at line 12 column 8
```

This includes:

- The file path (or `RAGENT_CONFIG_CONTENT` for the inline env var).
- The line and column from the serde error.
- A separator line.
- The exact source line that caused the failure.
- A caret (^) positioned under the error column.
- The underlying serde message.

The same format is used for `--config`, `RAGENT_CONFIG`, and `RAGENT_CONFIG_CONTENT` errors.

6. Backup and Restore

`Config::backup_global_config()` snapshots the current global `ragent.json` into a timestamped file inside `<config_dir>/ragent/saves/`. The backup name uses `YYYY-MM-DD.HH-MM-SS` (hyphens in the time portion for Windows NTFS compatibility). If multiple saves happen in the same second, a counter is appended. The write is atomic (temp-file-then-rename).

`Config::restore_global_config()` restores a backup over the global `ragent.json`. It validates that:

- The backup file exists and is a regular file.
- The resolved destination is exactly `<config_dir>/ragent.json`.
- The backup content is valid JSON.

The restore is also atomic.

These are used by the `/config backup` and `/config restore` slash commands.

7. Section Reference

7.1 username

Field	Type	Default	Description
username	Option<String>	None	Display name of the user, used in prompts and logs.

```
{ "username": "alice" }
```

7.2 defaultAgent

Field	Type	Default	Description
defaultAgent (alias default_agent)	String	"general"	Name of the agent used when none is specified via <code>--agent</code> .

```
{ "defaultAgent": "coder" }
```

The loader tracks whether the user explicitly set this field (`specified_default_agent`) so merge operations do not accidentally overwrite an explicit project-level choice with a global default.

7.3 provider

The `provider` section is the most important part of the config. It is a map keyed by provider id, where each entry configures one LLM provider.

```
{
  "provider": {
    "anthropic": {
      "env": ["ANTHROPIC_API_KEY"],
      "thinking": { "enabled": true, "level": "low" },
      "models": {
        "claude-sonnet-4-20250514": {
          "thinking": { "enabled": true, "level": "high", "budget_tokens":
16000 }
        }
      }
    }
  }
}
```

```

    },
    "generic_openai": {
      "env": ["GENERIC_OPENAI_API_KEY"],
      "api": { "base_url": "http://127.0.0.1:8080" }
    }
  }
}

```

ProviderConfig schema

Field	Type	Default	Description
env	Vec<String>	[]	Environment variable names required by this provider. ragent checks these are set before model discovery.
api	Option<ApiConfig>	None	Optional API endpoint and header overrides.
thinking	Option<ThinkingConfig>	None	Default thinking/reasoning config for all models under this provider. Overridden by per-model thinking .
models	HashMap<String, ModelConfig>	{}	Model definitions keyed by model id.
options	HashMap<String, Value>	{}	Arbitrary provider-specific options.

ApiConfig schema

Field	Type	Default	Description
base_url	Option<String>	None	Base URL for API requests, overriding the provider default.
headers	HashMap<String, String>	{}	Extra HTTP headers sent with every request.

ModelConfig schema

Field	Type	Default	Description
name	Option<String>	None	Human-readable display name.
cost	Option<Cost>	None	Per-token pricing (USD per million tokens).
capabilities	Option<Capabilities>	None	Feature flags for this model.
thinking	Option<ThinkingConfig>	None	Per-model thinking config; overrides provider-level default.

Cost schema

Field	Type	Description
input	f64	Cost per million input tokens (USD).
output	f64	Cost per million output tokens (USD).

Capabilities schema

Field	Type	Default	Description
reasoning	bool	false	Supports chain-of-thought reasoning.
streaming	bool	true	Supports streaming responses.
vision	bool	false	Can process image inputs.
tool_use	bool	true	Supports tool/function calling.
thinking_levels	Vec<ThinkingLevel>	[]	Which thinking/reasoning levels this model supports.

ThinkingConfig schema

Defined in `agent-types`, this is shared across providers.

Field	Type	Default	Description
enabled	bool	true	Whether thinking is enabled. When <code>false</code> , <code>level</code> is ignored.
level	ThinkingLevel	auto	Thinking depth: <code>auto</code> , <code>off</code> , <code>low</code> , <code>medium</code> , <code>high</code> .
budget_tokens	Option<u32>	None	Max tokens for thinking (Anthropic <code>budget_tokens</code>). <code>None</code> = provider default.
display	Option<ThinkingDisplay>	None	How thinking content is surfaced: <code>full</code> , <code>summarized</code> , <code>omitted</code> .

`ThinkingLevel` maps to provider-native parameters:

Provider	Native parameter
Anthropic	<code>thinking.type + effort / budget_tokens</code>
OpenAI / Copilot	<code>reasoning_effort</code> (<code>low</code> , <code>medium</code> , <code>high</code> , <code>none</code>)
Gemini	<code>thinkingConfig.thinkingLevel</code> (<code>minimal</code> , <code>low</code> , <code>medium</code> , <code>high</code> , <code>auto</code>)
Ollama	<code>think</code> boolean

Supported provider ids

Each provider is implemented in `crates/agent-llm/src/providers/`. The provider id (from `fn id() -> &'static str`) is the key in the `provider` map.

Provider id	Env var(s)	Notes
anthropic	ANTHROPIC_API_KEY	Claude models. Supports thinking/budget_tokens.
openai	OPENAI_API_KEY	GPT models. Supports reasoning_effort.
openai_responses	OPENAI_API_KEY	OpenAI Responses API variant.
gemini	GOOGLE_API_KEY or GEMINI_API_KEY	Google Gemini. Supports thinkingConfig.
ollama	OLLAMA_API_KEY (optional for local)	Local Ollama. <code>think</code> boolean for thinking.
ollama_cloud	OLLAMA_API_KEY	Ollama Cloud variant.
huggingface	HF_TOKEN or HUGGING_FACE_HUB_TOKEN	Hugging Face Inference.
copilot	GITHUB_COPILOT_TOKEN or GITHUB_TOKEN (via gh auth)	GitHub Copilot.
generic_openai	GENERIC_OPENAI_API_KEY or OPENAI_API_KEY; GENERIC_OPENAI_API_BASE for base URL	Any OpenAI-compatible endpoint.
azure_foundry	AZURE_AI_FOUNDRY_API_KEY; AZURE_AI_FOUNDRY_BASE for base URL	Azure AI Foundry.
azure_resource	(from <code>azureresources.json</code>)	Azure Resource (File) provider. Reads model definitions from a separate <code>azureresources.json</code> file.
bedrock	AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, AWS_REGION / AWS_BEDROCK_REGION, optional AWS_SESSION_TOKEN	Amazon Bedrock. Uses AWS SigV4.
xai	XAI_API_KEY; XAI_API_BASE for base URL	xAI (Grok).
router	(aggregates other providers)	Model Router virtual provider. Classifies prompts into tiers and routes to the best model.

Model Router configuration

The router is configured under `provider.router` and has its own `RouterConfig`:

```
{
  "provider": {
    "router": {
      "enabled": true,
      "tiers": {
```

```
    "SIMPLE":  { "models": [{ "provider": "ollama", "model": "llama3.2"
  }} },
    "MEDIUM": { "models": [{ "provider": "anthropic", "model": "claude-
sonnet-4-20250514" }] },
    "COMPLEX": { "models": [{ "provider": "anthropic", "model": "claude-
sonnet-4-20250514" }] },
    "REASONING": { "models": [{ "provider": "openai", "model": "o3" }] }
  },
  "boundaries": { "simple_medium": 0.25, "medium_complex": 0.50,
"complex_reasoning": 0.75 },
  "context_messages": 3,
  "default_timeout_ms": 30000
}
}
```

Field	Type	Default	Description
enabled	bool	false	Master switch. When false, requests pass through to the MEDIUM tier default without classification.
tiers	HashMap<String, TierConfig>	{}	Per-tier model fallback chains. Keys: SIMPLE, MEDIUM, COMPLEX, REASONING.
weights	WeightConfig	(15 dimensions, summing to 1.0)	Classifier dimension weights.
boundaries	BoundaryConfig	0.25 / 0.50 / 0.75	Score boundaries between tiers. Must be ascending and in [0.0, 1.0].
context_messages	usize	3	Number of recent conversation messages included in classification.
default_timeout_ms	u64	30000	Default request timeout.

TierConfig has models: Vec<TierEntry> (ordered fallback list) and optional timeout_ms. TierEntry has provider and model string fields.

The WeightConfig has 15 dimension weights (token_count, vocabulary_complexity, syntax_complexity, domain_specificity, ambiguity, context_dependency, reasoning_depth, creativity_level, emotional_complexity, multimodality, instruction_complexity, knowledge_recency, code_complexity, mathematical_complexity, image_attachment). Weights are normalised to sum to 1.0 at load time.

For more on providers and models, see the Tutorial (docs/howtos/tutorial.md).

7.4 agent

Per-agent configuration overrides, keyed by agent name. Applied on top of built-in agent defaults.

```
{
  "agent": {
    "coder": {
      "model": "anthropic/claude-sonnet-4-20250514",
      "temperature": 0.2,
      "max_steps": 200,
      "skills": ["rust-review"],
      "permission": [
        { "permission": "file:write", "pattern": "src/**", "action": "allow" }
      ]
    }
  }
}
```

AgentConfig schema

Field	Type	Default	Description
model	Option<String>	None	Model in "provider:model" format.
variant	Option<String>	None	Agent variant selector.
prompt	Option<String>	None	System prompt override.
temperature	Option<f32>	None	Sampling temperature override.
top_p	Option<f32>	None	Top-p (nucleus) sampling override.
mode	Option<String>	None	Agent mode: "primary", "subagent", or "all".
hidden	bool	false	Hide from user-facing listings.
permission	Vec<PermissionRule>	[]	Permission rules specific to this agent.
max_steps	Option<u32>	None	Maximum agentic loop iterations.
skills	Vec<String>	[]	Skill names to preload into the prompt.
options	HashMap<String, Value>	{}	Arbitrary agent-specific options.

For custom agents defined as JSON/Markdown files (OASF format), see [docs/howtos/custom-agents.md](#).

7.5 permission

Global permission rules applied to all agents. Rules are evaluated last-match-wins using glob patterns.

```
{
  "permission": [
    { "permission": "file:write", "pattern": "src/**", "action": "allow" },
```

```
{
  { "permission": "file:write", "pattern": "secrets/**", "action": "deny" },
  { "permission": "bash", "pattern": "git push --force", "action": "deny" },
  { "permission": "*", "pattern": "/etc/**", "action": "deny" }
}
```

PermissionRule schema

Field	Type	Description
permission	String (parsed to Permission)	The permission type. Supports flat names (read, bash) and namespaced categories (file:read, file:write).
pattern	Option<String>	Glob pattern matched against the resource path. None matches all.
action	PermissionAction	allow, deny, or ask.

Standard permission types

Name	Permission variant	Description
read / file:read	Read	Read access to files/resources.
edit / write / file:write	Edit	Write/edit access.
bash / execute	Bash	Shell command execution.
web / fetch	Web	Network/web access.
question	Question	Interactive question to user.
plan_enter / plan	PlanEnter	Enter planning phase.
plan_exit	PlanExit	Exit planning phase.
task	Task	Create/modify task items.
external_directory	ExternalDirectory	Access dirs outside project root.
doom_loop	DoomLoop	Detect/break infinite loops.
*	Custom("*")	Wildcard: applies to all permissions.
any other string	Custom(String)	User-defined permission type.

Evaluation order

- 1. **Permanent grants** (runtime "always allow" recorded via `PermissionChecker::record_always`) are checked first.
- 2. **Permission-specific rules** are evaluated last-match-wins.
- 3. **Wildcard rules** (`permission: "*"`) are evaluated last.
- 4. If no rule matches, the result is `ask`.

For the full permission system, YOLO mode, and autopilot, see `docs/howtos/permissions.md`.

7.6 command

User-defined slash-command shortcuts. Each entry maps a slash command name to a shell command.

```
{
  "command": {
    "build": { "command": "cargo build --release", "description": "Build
release binary" },
    "test": { "command": "cargo test -- --nocapture", "description": "Run all
tests with output" }
  }
}
```

CommandDef schema

Field	Type	Description
command	String	Shell command to execute.
description	String	Human-readable description shown in help output.

7.7 mcp

MCP (Model Context Protocol) server definitions, keyed by server id.

```
{
  "mcp": {
    "filesystem": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem",
"/home/user/docs"],
      "env": {}
    },
    "remote-api": {
      "type": "sse",
      "url": "https://mcp.example.com/sse",
      "headers": { "Authorization": "Bearer token123" },
      "disabled": false,
      "notification": "inject_summary"
    }
  }
}
```

McpServerConfig schema

Field	Type	Default	Description
type	McpTransport	stdio	Transport: <code>stdio</code> , <code>sse</code> , or <code>http</code> .
command	Option<String>	None	Executable path/name (stdio).
args	Vec<String>	[]	Command-line arguments (stdio).
env	HashMap<String, String>	{}	Environment variables (stdio).
url	Option<String>	None	URL endpoint (SSE/HTTP).
headers	HashMap<String, String>	{}	HTTP headers (SSE/HTTP).
disabled	bool	false	If <code>true</code> , server is configured but not started.
notification	McpNotificationMode	none	Push notification handling: <code>none</code> , <code>inject_summary</code> , <code>inject_and_run</code> .

7.8 instructions

Additional instruction strings appended to agent system prompts.

```
{
  "instructions": [
    "Always use rustfmt before writing Rust files.",
    "Prefer iterator adapters over manual loops."
  ]
}
```

Type: `Vec<String>`. Default: `[]`. These are appended to every agent's system prompt, merged (union) across config layers.

7.9 skill_dirs

Additional directories to scan for skill definitions, beyond the built-in locations.

```
{
  "skill_dirs": ["~/my-skills", "/shared/team-skills"]
}
```

Type: `Vec<String>`. Default: `[]`. Merged (union) across config layers.

7.10 experimental

Feature flags for experimental functionality.

```
{
  "experimental": {
    "open_telemetry": false,
    "parallel_tool_calls": false,
    "max_background_agents": 8,
    "background_agent_timeout": 3600
  }
}
```

ExperimentalFlags schema

Field	Type	Default	Description
open_telemetry	bool	false	Enable OpenTelemetry trace export (legacy; prefer <code>telemetry.otel</code>).
parallel_tool_calls	bool	false	Allow multiple tool calls from a single model turn to execute in parallel.
max_background_agents	usize	8	Maximum concurrent background sub-agent tasks.
background_agent_timeout	u64	3600	Timeout in seconds for background sub-agent tasks.

`open_telemetry` is mapped as a deprecated alias: if `telemetry.otel` is not explicitly enabled but `experimental.open_telemetry` is `true`, telemetry is enabled with default settings and a deprecation warning is emitted.

7.11 hooks

Lifecycle hooks. Currently a placeholder (`Vec<serde_json::Value>`). See `docs/howtos/hooks.md` for the hooks system documentation.

Type: `Vec<Value>`. Default: `[]`. Merged (append) across config layers.

7.12 bash

User-defined bash command allowlist and denylist additions.

```
{
  "bash": {
    "allowlist": ["curl", "wget"],
    "denylist": ["git push --force", "rm -rf /"]
  }
}
```

BashConfig schema

Field	Type	Default	Description
allowlist	Vec<String>	[]	Command prefixes exempted from the banned-command check (e.g. "curl").
denylist	Vec<String>	[]	Substring patterns that always reject a command (e.g. "git push --force").

Both lists are unioned (deduplicated) across global and project configs. The `/bash add` and `/bash remove` slash commands mutate these lists and persist changes to the targeted config file.

Allowlist entries match the first word of the command. **Denylist** entries are substring-matched anywhere in the command string.

This interacts with the 7-layer bash security model. See [docs/howtos/permissions.md](#) for the full model.

7.13 dirs

User-defined directory/file path allowlist and denylist additions.

```
{
  "dirs": {
    "allowlist": ["src/**", "tests/**", "*.rs"],
    "denylist": ["secrets/**", "/etc/**", ".env"]
  }
}
```

DirsConfig schema

Field	Type	Default	Description
allowlist	Vec<String>	[]	Glob patterns for paths automatically allowed (no prompt).
denylist	Vec<String>	[]	Glob patterns unconditionally rejected.

Both lists are unioned across config layers. **Denylist takes precedence over allowlist.** The `/dirs add` and `/dirs remove` slash commands mutate these lists.

Built-in denylist patterns (always active, not configurable) include system directories: `/bin/**`, `/sbin/**`, `/boot/**`, `/dev/**`, `/proc/**`, `/sys/**`, `/etc/**`, `/usr/bin/**`, `/usr/sbin/**`, `/usr/lib/**`, `/lib/**`, `/lib64/**`, and their macOS/Windows equivalents.

7.14 tool_visibility

Controls which tool families are advertised to the LLM. When a switch is `false`, all tools in that family are hidden from tool definitions and system-prompt listings. Hidden tools remain registered and executable.

```
{
  "tool_visibility": {
```

```
    "office": true,
    "github": true,
    "gitlab": false,
    "teams": true,
    "agents": true,
    "plan": true,
    "codeindex": true,
    "masterfetch": true,
    "browser": true,
    "finance": true
  }
}
```

Switches and defaults

Switch	Default	Tools governed
office	false	office_read, office_write, office_info, libre_read, libre_write, libre_info, pdf_read, pdf_write
github	false	github_list_issues, github_get_issue, github_create_issue, github_comment_issue, github_close_issue, github_list_prs, github_get_pr, github_create_pr, github_merge_pr, github_review_pr
gitlab	false	19 GitLab tools (issues, MRs, pipelines, jobs)
teams	false	20 team coordination tools
agents	false	cancel_agent, list_agents, new_agent, agent_complete, wait_agents
plan	false	plan_enter, plan_exit
codeindex	true	codeindex_search, codeindex_status, codeindex_symbols, codeindex_references, codeindex_dependencies, codeindex_reindex
masterfetch	true	mf_fetch, mf_crawl, mf_search, mf_screenshot, mf_cache_clear, mf_version
browser	true	browser
finance	true	stock_quote, stock_history, stock_fundamentals, currency_rate, currency_history, stock_search, stock_options

The `codeindex`, `masterfetch`, `browser`, and `finance` switches are serialised **only** when explicitly set (tracked by `specified` flags). This lets the default config omit the key so code-level default changes propagate, while ensuring an explicit user toggle persists on save.

The `/tools` slash command toggles these at runtime. See `docs/howtos/tool-visibility.md` and `docs/howtos/tools.md`.

7.15 code_index

Persistent configuration for the codebase indexing subsystem.

```
{
  "code_index": {
    "enabled": true,
    "max_file_size": 1048576,
    "extra_exclude_dirs": ["vendor", "node_modules"],
    "extra_exclude_patterns": ["*.generated.rs"]
  }
}
```

CodeIndexConfig schema

Field	Type	Default	Description
enabled	bool	true	Whether code indexing is enabled.
max_file_size	u64	1048576 (1 MB)	Maximum file size in bytes to index.
extra_exclude_dirs	Vec<String>	[]	Additional directory names to exclude from scanning.
extra_exclude_patterns	Vec<String>	[]	Additional glob patterns to exclude.

The `enabled` field is serialised only when explicitly set (tracked by `specified.enabled`), so `/codeindex on//codeindex off` toggles persist on save.

For the full code index guide, see [docs/howtos/codeindex.md](#).

7.16 stream

LLM streaming behaviour: timeouts and retries.

```
{
  "stream": {
    "initial_response_timeout_secs": 300,
    "timeout_secs": 120,
    "max_retries": 4,
    "retry_backoff_secs": 2
  }
}
```

StreamConfig schema

Field	Type	Default	Validation	Description
<code>initial_response_timeout_secs</code>	<code>u64</code>	<code>300</code>	<code>>= 5, >= timeout_secs</code>	Seconds the HTTP client waits for the first byte of a streaming response.
<code>timeout_secs</code>	<code>u64</code>	<code>120</code>	<code>>= 5</code>	Seconds of silence between stream deltas before a stream is considered stalled.
<code>max_retries</code>	<code>u32</code>	<code>4</code>	<code><= 32</code>	Maximum retry attempts after a stall or connection failure.
<code>retry_backoff_secs</code>	<code>u64</code>	<code>2</code>	<code>—</code>	Backoff multiplier per retry. Attempt N waits <code>N * retry_backoff_secs</code> seconds.

`validate()` returns a list of problems; the session processor refuses to start if any are present.

7.17 `memory`

Memory system configuration: blocks, structured store, semantic search, retrieval, auto-extraction, decay, and cross-project sharing.

```
{
  "memory": {
    "enabled": true,
    "tier": "core",
    "structured": { "enabled": true },
    "semantic": {
      "enabled": false,
      "model": "all-MiniLM-L6-v2",
      "dimensions": 384
    },
    "retrieval": {
      "max_memories_per_prompt": 5,
      "recency_weight": 0.3,
      "relevance_weight": 0.7
    },
    "auto_extract": {
      "enabled": false,
      "require_confirmation": true
    },
    "decay": {
      "factor": 0.95,
      "min_confidence": 0.1
    },
    "cross_project": {
      "enabled": false,
```

```
        "search_global": true,
        "project_override": true
    }
}
```

MemoryConfig schema

Field	Type	Default	Description
enabled	bool	true	Master switch for the memory system.
tier	String	"core"	Memory tier: "core" (file blocks), "structured" (SQLite store), "semantic" (with embeddings).
structured	StructuredMemoryConfig	{ enabled: true }	Structured store configuration.
retrieval	RetrievalConfig	(see below)	Retrieval configuration for prompt injection.
semantic	SemanticConfig	(see below)	Semantic search (embedding) configuration.
auto_extract	AutoExtractConfig	(see below)	Automatic memory extraction.
decay	DecayConfig	(see below)	Confidence decay.
cross_project	CrossProjectConfig	(see below)	Cross-project memory sharing.

Sub-schemas

StructuredMemoryConfig: enabled: bool (default true).

RetrievalConfig:

Field	Type	Default	Description
max_memories_per_prompt	usize	5	Max structured memories injected into the system prompt.
recency_weight	f64	0.3	Weight for recency when ranking (0.0-1.0).
relevance_weight	f64	0.7	Weight for relevance when ranking (0.0-1.0).

SemanticConfig:

Field	Type	Default	Description
enabled	bool	false	Whether semantic search via embeddings is enabled. Requires the embeddings Cargo feature.
model	String	"all-MiniLM-L6-v2"	ONNX sentence-transformer model name.

Field	Type	Default	Description
dimensions	usize	384	Embedding vector dimensions (must match model output).

AutoExtractConfig:

Field	Type	Default	Description
enabled	bool	false	Whether automatic memory extraction is enabled.
require_confirmation	bool	true	Whether extracted candidates require explicit confirmation before storage.

DecayConfig:

Field	Type	Default	Description
factor	f64	0.95	Multiplicative decay factor per day since last access. 1.0 disables decay.
min_confidence	f64	0.1	Minimum confidence floor. Memories never decay below this.

CrossProjectConfig:

Field	Type	Default	Description
enabled	bool	false	Whether cross-project memory sharing is enabled.
search_global	bool	true	Whether search includes global memories.
project_override	bool	true	Whether project-specific blocks override global blocks with the same label.

7.18 compaction

Context-window compaction configuration (OpenCode-derived summarisation).

```
{
  "compaction": {
    "auto": true,
    "threshold": 0.7,
    "buffer": 0.10,
    "keep": { "tokens": 0.20 }
  }
}
```

CompactionConfig schema

Field	Type	Default	Description
auto	bool	true	Whether automatic pre-send compaction is enabled. When false, only provider context-overflow errors

Field	Type	Default	Description
			trigger emergency compaction.
threshold	Option<f64>	Some(0.7)	Fraction of context window at which to trigger compaction (0.0-1.0). None falls back to buffer-based trigger. Clamped to minimum 0.7 so routine prompts under 70% never trigger compaction.
buffer	f64	0.10	Token buffer as a fraction of context window. When threshold is None , triggers when tokens exceed <code>context_window - max(output_tokens, context_window * buffer)</code> .
keep	KeepConfig	{ tokens: Some(0.20) }	Recent turns kept verbatim after compaction.

KeepConfig: `tokens: Option<f64>` (default `Some(0.20)`) — max fraction of context window preserved as the verbatim tail.

Fixed constants:

- `summary_output_tokens()`: 4096 tokens requested for a compaction summary.
- `tool_output_max_chars()`: 2000 chars — long tool outputs are truncated before being serialised into the compaction prompt.

7.19 gitlab

GitLab integration configuration. Values set here override those stored in the ragent database (set via `/gitlab setup`). Environment variables take the highest priority.

```
{
  "gitlab": {
    "instance_url": "https://gitlab.example.com",
    "token": "glpat-xxxxxxxxxxxx",
    "username": "myuser"
  }
}
```

GitLabIntegrationConfig schema

Field	Type	Default	Description
instance_url	Option<String>	None	GitLab instance base URL.
token	Option<String>	None	Personal Access Token.
username	Option<String>	None	GitLab username / identity.

Env var overrides: `GITLAB_TOKEN`, `GITLAB_URL`, `GITLAB_USERNAME`.

7.20 hidden_tools

Tool names to hide from the LLM. Hidden tools remain registered and executable; they are simply not advertised to the model.

```
{
  "hidden_tools": ["github_list_issues", "github_get_issue",
"gitlab_list_mrs"]
}
```

Type: `Vec<String>`. Default: `[]`. Unioned (deduplicated) across config layers.

The `effective_hidden_tools()` method also computes tools hidden by `tool_visibility` switches set to `false`.

7.21 yolo

YOLO mode bypasses all command validation and tool restrictions.

```
{ "yolo": false }
```

Type: `bool`. Default: `false`.

When enabled, the following safety checks are skipped:

- Bash denied patterns (destructive commands like `rm -rf /` are allowed)
- Dynamic context allowlist (any executable can run in skill bodies)
- MCP config validation (shell metacharacters and unvalidated paths permitted)

Warning: This is inherently dangerous. Use only when you trust the agent and its inputs completely, or for local development/debugging.

The `Alt+Y` keybinding and `/yolo` slash command toggle this at runtime and persist the change. See <docs/howtos/permissions.md>.

7.22 edit_log

Edit-operation logging for `edit` and `multi_edit` tools.

```
{ "edit_log": false }
```

Type: `bool`. Default: `false`.

When enabled, all edit operations are logged. The `Alt+E` keybinding and `/editlog` slash command toggle this at runtime and persist the change.

7.23 prices

User-defined price overrides for cost estimation. Each entry overrides the built-in price table for a specific model. Prices are in USD per 1,000,000 tokens.

```
{
  "prices": [
    { "model": "gpt-4o", "input_per_1m": 2.50, "output_per_1m": 10.00 },
    { "model": "claude-sonnet-4-20250514", "input_per_1m": 3.00,
      "output_per_1m": 15.00 }
  ]
}
```

PriceEntry schema

Field	Type	Description
model	String	Model identifier as returned by the provider.
input_per_1m	f64	Price per 1M input/prompt tokens (USD).
output_per_1m	f64	Price per 1M output/completion tokens (USD).

When merged, overlay entries replace base entries with the same model id (last-wins per model).

7.24 browser

Browser automation configuration (Chrome DevTools Protocol endpoint).

```
{
  "browser": {
    "cdp_endpoint": "http://127.0.0.1:9222",
    "default_headless": true
  }
}
```

BrowserConfig schema

Field	Type	Default	Description
cdp_endpoint	Option<String>	None	CDP HTTP endpoint URL. When None /empty, defaults to <code>http://127.0.0.1:9222</code> .
default_headless	bool	true	Default headless mode for the setup action.

7.25 channels

External messaging channel configuration for the `send_channel_message` tool.

```
{
  "channels": {
    "enabled": true,
    "telegram": { "bot_token": "env:TELEGRAM_BOT_TOKEN", "chat_id": "-100123" },
    "discord": { "webhook_url": "https://discord.com/api/webhooks/..." }
  }
}
```

ChannelsConfig schema

Field	Type	Default	Description
enabled	bool	false	Master switch for the channel messaging tool.
telegram	Option<TelegramChannelConfig>	None	Telegram bot channel.
discord	Option<DiscordChannelConfig>	None	Discord webhook channel.

TelegramChannelConfig: bot_token: Option<String>, chat_id: Option<String>, base_url: Option<String> (defaults to https://api.telegram.org).

DiscordChannelConfig: webhook_url: Option<String>.

All token/secret fields support the `env:VAR_NAME` indirection — the value is read from the named environment variable at use time, so secrets do not need to live in the config file.

7.26 gmail

Gmail tool configuration (OAuth2 client credentials).

```
{
  "gmail": {
    "client_id": "...apps.googleusercontent.com",
    "client_secret": "env:GMAIL_CLIENT_SECRET"
  }
}
```

GmailConfig schema

Field	Type	Default	Description
client_id	Option<String>	None	OAuth2 client ID. Supports <code>env:VAR_NAME</code> . Falls back to <code>GMAIL_CLIENT_ID</code> .
client_secret	Option<String>	None	OAuth2 client secret. Supports <code>env:VAR_NAME</code> . Falls back to <code>GMAIL_CLIENT_SECRET</code> .

Field	Type	Default	Description
<code>base_url</code>	<code>Option<String></code>	<code>None</code>	Optional HTTP endpoint override (defaults to <code>https://gmail.googleapis.com</code>).

The OAuth2 access/refresh tokens are managed by the `gmail` tool itself (`auth/status/logout` actions) and stored encrypted in `ragent-storage` — never in this file.

7.27 `telemetry`

OpenTelemetry metrics export configuration.

```
{
  "telemetry": {
    "otel": {
      "enabled": false,
      "endpoint": "http://localhost:4318",
      "protocol": "http",
      "export_interval_seconds": 30,
      "export_timeout_seconds": 10,
      "service_name": "ragent",
      "resource_attributes": { "deployment.environment": "production" },
      "metrics": { "ragent.tool.call_count": false },
      "internal_port": null,
      "cardinality_limit": 1000
    }
  }
}
```

`TelemetryConfig` schema

Wraps `OtelConfig` as `telemetry.otel`.

`OtelConfig` schema

Field	Type	Default	Description
<code>enabled</code>	<code>bool</code>	<code>false</code>	Master on/off switch.
<code>endpoint</code>	<code>String</code>	<code>"http://localhost:4318"</code>	OTLP endpoint base URL. Must be <code>http://</code> or <code>https://</code> .
<code>protocol</code>	<code>OtelProtocol</code>	<code>http</code>	Transport: <code>http</code> or <code>grpc</code> .
<code>export_interval_seconds</code>	<code>u64</code>	<code>30</code>	Batch export interval. Must be

Field	Type	Default	Description
			> 0 when enabled.
export_timeout_seconds	u64	10	Per-export request timeout. Must be > 0 when enabled.
service_name	String	"ragent"	service.name resource attribute. Must not be empty.
resource_attributes	HashMap<String, String>	{}	Custom resource attributes appended to every metric.
metrics	HashMap<String, bool>	{}	Per-metric enable/disable toggles. Absent = enabled by default.
internal_port	Option<u16>	None	Optional in-process Prometheus text endpoint port.
cardinality_limit	usize	1000	Max distinct attribute combinations per metric before overflow into unknown bucket.

The legacy `experimental.open_telemetry` flag is mapped as a deprecated alias via `apply_legacy_flag()`.

7.28 agent_perf

Agent-loop performance configuration.

```
{
  "agent_perf": {
    "enabled": true,
    "profiling": false,
    "step_budget_secs": 300,
```

```
    "stall_timeout_secs": 60,
    "max_concurrent_tools": 4,
    "parallel_independent_tools": true
  }
}
```

AgentPerfConfig schema

Field	Type	Default	Validation	Description
enabled	bool	true	—	Master switch. When false , all performance optimisations short-circuit.
profiling	bool	false	—	Emit detailed per-scope timing logs at info level.
step_budget_secs	u64	300	>= 5	Maximum wall-clock seconds per agent step.
stall_timeout_secs	u64	60	>= 5	Maximum seconds without a stream delta before stall recovery fires.
max_concurrent_tools	u32	min(available_parallelism, 4)	>= 1	Maximum parallel tool calls per turn.
parallel_independent_tools	bool	true	—	Execute independent tool calls in parallel.

`validate()` returns a list of problems; the agent loop refuses to start if any are present.

7.29 finance

Paid finance-provider configuration for the stock/currency toolset.

```
{
  "finance": {
    "provider": "alpha_vantage",
    "api_key": "env:ALPHA_VANTAGE_API_KEY",
    "requests_per_minute": 5,
    "min_call_interval_seconds": 5,
    "yahoo_fallback": true
  }
}
```

FinanceProviderConfig schema

Field	Type	Default	Description
provider	String	"yahoo"	Selected provider: "yahoo" (free, default) or "alpha_vantage".
api_key	Option<String>	None	API key when a paid provider is selected.
base_url	Option<String>	None	Optional base URL override for the paid provider.
requests_per_minute	Option<u32>	None	Optional request rate limit.
user_agent	Option<String>	None	Optional custom User-Agent header for the free Yahoo provider.
min_call_interval_seconds	u64	5	Minimum seconds between any two finance API calls.
yahoo_fallback	Option<bool>	None	Whether to fall back to Yahoo when the paid provider fails. None = enabled for Yahoo, disabled for paid providers.

7.30 Search API keys

Several API keys for the **mf_search** / **websearch** tools can be stored in the config file. Environment variables always take precedence.

Field	Type	Default	Env var override	Description
tavily_api_key	Option<String>	None	TAVILY_API_KEY	Tavily search API key for the websearch tool.
langsearch_api_key	Option<String>	None	—	LangSearch API key for mf_search . Masked in diagnostics.
perplexity_api_key	Option<String>	None	—	Perplexity Sonar API key for mf_search . Masked

Field	Type	Default	Env var override	Description
				in diagnostics.
<code>exa_api_key</code>	<code>Option<String></code>	None	<code>EXA_API_KEY</code>	Exa Search API key for <code>mf_search</code> . Masked in diagnostics.
<code>openalex_email</code>	<code>Option<String></code>	None	<code>OPENALEX_EMAIL</code>	OpenAlex polite-pool email for <code>mf_search</code> . Masked in diagnostics.

When present, `mf_search` queries the corresponding backend as an additional search engine alongside the keyless backends (DuckDuckGo, Brave, OpenAlex, Wikipedia).

7.31 `sdd`

Spec-Driven Development (SDD) capability toggles. All flags default to `false` (opt-in).

```
{
  "sdd": {
    "clarification_markers": true,
    "quality_checklists": true,
    "constitution": true,
    "phase_minus_one_gates": true,
    "branch_per_spec": true,
    "research_artifacts": true,
    "data_model": true,
    "contracts": true,
    "quickstart": true,
    "test_first_ordering": true,
    "consistency_checks": true
  }
}
```

`SddConfig` schema

Field	Type	Default	Description
<code>clarification_markers</code>	<code>bool</code>	<code>false</code>	Enable <code>[NEEDS CLARIFICATION]</code> marker detection (FR-002).
<code>quality_checklists</code>	<code>bool</code>	<code>false</code>	Embed quality checklists in spec/plan templates (FR-006).
<code>constitution</code>	<code>bool</code>	<code>false</code>	Generate and parse <code>CONSTITUTION.md</code> (FR-007).
<code>phase_minus_one_gates</code>	<code>bool</code>	<code>false</code>	Enable Phase -1 pre-implementation gate validation (FR-008).
<code>branch_per_spec</code>	<code>bool</code>	<code>false</code>	Create a git branch per spec (FR-009).
<code>research_artifacts</code>	<code>bool</code>	<code>false</code>	Link research artifacts into SPEC.md frontmatter (FR-010).

Field	Type	Default	Description
<code>data_model</code>	<code>bool</code>	<code>false</code>	Generate <code>data-model.md</code> during <code>/spec plan</code> (FR-011).
<code>contracts</code>	<code>bool</code>	<code>false</code>	Generate <code>contracts/</code> directory during <code>/spec plan</code> (FR-012).
<code>quickstart</code>	<code>bool</code>	<code>false</code>	Generate <code>quickstart.md</code> validation scenarios (FR-013).
<code>test_first_ordering</code>	<code>bool</code>	<code>false</code>	Enforce test-first file creation ordering in plans (FR-014).
<code>consistency_checks</code>	<code>bool</code>	<code>false</code>	Run ambiguity, contradiction, and gap consistency checks (FR-015).

Merge uses OR semantics: a flag enabled in either base or overlay stays enabled.

For the spec system, see [docs/howtos/spec.md](#).

7.32 trigger

Dynamic trigger rule system configuration.

```
{
  "trigger": {
    "enabled": true,
    "poll_interval_secs": 30,
    "max_rules": 32
  }
}
```

TriggerConfig schema

Field	Type	Default	Description
<code>enabled</code>	<code>bool</code>	<code>true</code>	Master feature gate. When <code>false</code> , all trigger functionality no-ops.
<code>poll_interval_secs</code>	<code>u64</code>	<code>30</code>	Interval at which dynamic trigger rules poll their conditions.
<code>max_rules</code>	<code>usize</code>	<code>32</code>	Maximum number of dynamic trigger rules per session.

`is_empty()` returns `true` when all fields are at default values, so the section is omitted from serialised output when nothing is configured.

7.33 piegap

Pie feature gap toggles. Each flag gates a standalone pie-derived feature. All default to `false` (opt-in).

```
{
  "piegap": {
    "triggers": true,
```

```
    "mcp_notifications": true,
    "inbox": true,
    "hooks": true,
    "archive": true,
    "bug_report": true,
    "templates": true,
    "goal": true,
    "web_ui": true,
    "undo": true,
    "session_naming": true
  }
}
```

PieGapConfig schema

Field	Type	Default	Description
triggers	bool	false	Enable dynamic trigger rules (G-01).
mcp_notifications	bool	false	Enable MCP notification push events (G-02).
inbox	bool	false	Enable stateful loops + triage inbox (G-03).
hooks	bool	false	Enable lifecycle hooks (G-04).
archive	bool	false	Enable portable session archive export/import (G-05).
bug_report	bool	false	Enable bug report generation (G-06).
templates	bool	false	Enable reusable prompt templates (G-07).
goal	bool	false	Enable goal-based autonomous stop hook (G-10).
web_ui	bool	false	Enable browser-based web UI (G-12).
undo	bool	false	Enable <code>/undo</code> slash command (G-13).
session_naming	bool	false	Enable session naming (G-14).

Merge uses OR semantics. `is_empty()` returns `true` when no flags are enabled.

7.34 research

Research subsystem configuration.

```
{
  "research": {
    "open_access_recovery": true,
    "contact_email": "user@example.com",
    "oa_min_full_text_chars": 1000
  }
}
```

ResearchConfig schema

Field	Type	Default	Description
open_access_recovery	bool	false	Enable open-access recovery via Unpaywall and Europe PMC for short scholarly sources (FR-011).
contact_email	Option<String>	None	Contact email required by Unpaywall's terms of service (FR-012).
oa_min_full_text_chars	usize	1000	Minimum full-text length (chars) that triggers OA recovery.

`open_access_recovery` uses OR semantics on merge. `contact_email` and `oa_min_full_text_chars` override base when present. `is_empty()` returns `true` when at default values.

For the research system, see `docs/howtos/research.md`.

8. Full Example File

The following is a comprehensive example showing every section. You do not need all of these — every section has defaults, so an empty `{}` is valid.

```
{
  "username": "alice",
  "defaultAgent": "coder",

  "provider": {
    "anthropic": {
      "env": ["ANTHROPIC_API_KEY"],
      "thinking": { "enabled": true, "level": "low" },
      "models": {
        "claude-sonnet-4-20250514": {
          "thinking": {
            "enabled": true,
            "level": "high",
            "budget_tokens": 16000
          }
        }
      }
    },
  },
  "openai": {
    "env": ["OPENAI_API_KEY"]
  },
  "generic_openai": {
    "env": ["GENERIC_OPENAI_API_KEY"],
    "api": { "base_url": "http://127.0.0.1:8080" }
  },
  "router": {
    "enabled": true,
    "tiers": {
      "SIMPLE": {
```

```

    "models": [{ "provider": "ollama", "model": "llama3.2" }]
  },
  "MEDIUM": {
    "models": [
      { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
    ]
  },
  "COMPLEX": {
    "models": [
      { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
    ]
  },
  "REASONING": {
    "models": [{ "provider": "openai", "model": "o3" }]
  }
}
},
"agent": {
  "coder": {
    "model": "anthropic/claude-sonnet-4-20250514",
    "temperature": 0.2,
    "max_steps": 200,
    "skills": ["rust-review"]
  }
},
"permission": [
  { "permission": "file:write", "pattern": "src/**", "action": "allow" },
  { "permission": "file:write", "pattern": "secrets/**", "action": "deny" },
  { "permission": "bash", "pattern": "git push --force", "action": "deny" }
],
"command": {
  "build": {
    "command": "cargo build --release",
    "description": "Build release binary"
  }
},
"mcp": {
  "filesystem": {
    "type": "stdio",
    "command": "npx",
    "args": ["-y", "@modelcontextprotocol/server-filesystem",
"/home/user/docs"]
  }
},
"instructions": [
  "Always use rustfmt before writing Rust files."
],
"skill_dirs": ["~/my-skills"],

```

```
"experimental": {
  "parallel_tool_calls": false,
  "max_background_agents": 8,
  "background_agent_timeout": 3600
},

"bash": {
  "allowlist": ["curl", "wget"],
  "denylist": ["git push --force"]
},

"dircs": {
  "allowlist": ["src/**", "tests/**"],
  "denylist": ["secrets/**", ".env"]
},

"tool_visibility": {
  "office": true,
  "github": true,
  "gitlab": false,
  "teams": true,
  "agents": true,
  "plan": true
},

"code_index": {
  "enabled": true,
  "max_file_size": 1048576,
  "extra_exclude_dirs": ["vendor"]
},

"stream": {
  "initial_response_timeout_secs": 300,
  "timeout_secs": 120,
  "max_retries": 4,
  "retry_backoff_secs": 2
},

"memory": {
  "enabled": true,
  "tier": "core",
  "structured": { "enabled": true },
  "semantic": {
    "enabled": false,
    "model": "all-MiniLM-L6-v2",
    "dimensions": 384
  },
  "retrieval": {
    "max_memories_per_prompt": 5,
    "recency_weight": 0.3,
    "relevance_weight": 0.7
  },
  "auto_extract": {
    "enabled": false,
```

```
    "require_confirmation": true
  },
  "decay": {
    "factor": 0.95,
    "min_confidence": 0.1
  },
  "cross_project": {
    "enabled": false,
    "search_global": true,
    "project_override": true
  }
},

"compaction": {
  "auto": true,
  "threshold": 0.7,
  "buffer": 0.10,
  "keep": { "tokens": 0.20 }
},

"gitlab": {
  "instance_url": "https://gitlab.example.com",
  "token": "env:GITLAB_TOKEN",
  "username": "myuser"
},

"hidden_tools": [],

"yolo": false,
"edit_log": false,

"prices": [
  {
    "model": "gpt-4o",
    "input_per_1m": 2.50,
    "output_per_1m": 10.00
  }
],

"browser": {
  "cdp_endpoint": "http://127.0.0.1:9222",
  "default_headless": true
},

"channels": {
  "enabled": true,
  "telegram": {
    "bot_token": "env:TELEGRAM_BOT_TOKEN",
    "chat_id": "-100123"
  }
},

"gmail": {
  "client_id": "env:GMAIL_CLIENT_ID",
  "client_secret": "env:GMAIL_CLIENT_SECRET"
```

```
},

"telemetry": {
  "otel": {
    "enabled": false,
    "endpoint": "http://localhost:4318",
    "protocol": "http",
    "export_interval_seconds": 30,
    "export_timeout_seconds": 10,
    "service_name": "ragent",
    "cardinality_limit": 1000
  }
},

"agent_perf": {
  "enabled": true,
  "step_budget_secs": 300,
  "stall_timeout_secs": 60,
  "max_concurrent_tools": 4,
  "parallel_independent_tools": true
},

"finance": {
  "provider": "yahoo",
  "min_call_interval_seconds": 5
},

"tavily_api_key": null,
"langsearch_api_key": null,
"perplexity_api_key": null,
"exa_api_key": null,
"openalex_email": null,

"sdd": {
  "clarification_markers": false,
  "quality_checklists": false
},

"trigger": {
  "enabled": true,
  "poll_interval_secs": 30,
  "max_rules": 32
},

"piegap": {
  "triggers": false,
  "hooks": false,
  "undo": false
},

"research": {
  "open_access_recovery": false,
  "contact_email": null,
  "oa_min_full_text_chars": 1000
}
```

```
}
}
```

9. Common Recipes

Minimal config (just an API key reference)

```
{
  "provider": {
    "anthropic": { "env": ["ANTHROPIC_API_KEY"] }
  }
}
```

The API key itself is read from the environment variable; it does not need to be in the config file.

Local-first (Ollama only)

```
{
  "defaultAgent": "coder",
  "provider": {
    "ollama": {}
  }
}
```

ragent resolves to the first available local/self-hosted provider when no model is explicitly configured.

Custom OpenAI-compatible endpoint

```
{
  "provider": {
    "generic_openai": {
      "env": ["GENERIC_OPENAI_API_KEY"],
      "api": { "base_url": "http://127.0.0.1:8080" }
    }
  }
}
```

The base URL can also be set via `GENERIC_OPENAI_API_BASE` env var.

Allow writes to src/, deny secrets

```
{
  "permission": [
    { "permission": "file:write", "pattern": "src/**", "action": "allow" },
    { "permission": "file:write", "pattern": "secrets/**", "action": "deny" }
  ]
}
```

```
]
}
```

Enable GitHub and teams tools

```
{
  "tool_visibility": {
    "github": true,
    "teams": true
  }
}
```

Disable code indexing

```
{
  "code_index": { "enabled": false }
}
```

Semantic memory with embeddings

```
{
  "memory": {
    "semantic": {
      "enabled": true,
      "model": "all-MiniLM-L6-v2",
      "dimensions": 384
    }
  }
}
```

Requires the [embeddings](#) Cargo feature. When disabled, memory search falls back to FTS5-only mode.

OpenTelemetry export to a collector

```
{
  "telemetry": {
    "otel": {
      "enabled": true,
      "endpoint": "http://otel-collector:4318",
      "protocol": "http",
      "export_interval_seconds": 15,
      "service_name": "ragent-prod",
      "resource_attributes": {
        "deployment.environment": "production",
        "host.name": "build-server-01"
      }
    }
  }
}
```

```
}  
}  
}
```

Telegram notifications

```
{  
  "channels": {  
    "enabled": true,  
    "telegram": {  
      "bot_token": "env:TELEGRAM_BOT_TOKEN",  
      "chat_id": "env:TELEGRAM_CHAT_ID"  
    }  
  }  
}
```

Model Router with fallback chains

```

{
  "provider": {
    "router": {
      "enabled": true,
      "tiers": {
        "SIMPLE": {
          "models": [
            { "provider": "ollama", "model": "llama3.2" },
            { "provider": "openai", "model": "gpt-4o-mini" }
          ]
        },
        "MEDIUM": {
          "models": [
            { "provider": "anthropic", "model": "claude-sonnet-4-20250514" },
            { "provider": "openai", "model": "gpt-4o" }
          ]
        },
        "COMPLEX": {
          "models": [
            { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
          ]
        },
        "REASONING": {
          "models": [
            { "provider": "openai", "model": "o3" },
            { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
          ]
        }
      }
    }
  }
}

```

Override pricing for a custom model

```

{
  "prices": [
    {
      "model": "my-custom-model",
      "input_per_1m": 1.00,
      "output_per_1m": 5.00
    }
  ]
}

```

Inline config via environment variable

```
RAGENT_CONFIG_CONTENT='{"defaultAgent":"ask","yolo":true}' ragent run "hello"
```

This is merged last, with the highest precedence.

10. Related Documents

Document	Topic
docs/howtos/tutorial.md	End-to-end tutorial from setup to release
docs/howtos/permissions.md	Permissions, autopilot, and YOLO mode
docs/howtos/tools.md	Complete tool reference
docs/howtos/tool-visibility.md	Tool family visibility toggles
docs/howtos/custom-agents.md	Custom agent definitions (OASF / Markdown)
docs/howtos/teams.md	Teams and swarm coordination
docs/howtos/hooks.md	Lifecycle hooks
docs/howtos/codeindex.md	Code index (tree-sitter, search, graph)
docs/howtos/spec.md	Spec lifecycle management
docs/howtos/research.md	Research system
docs/howtos/finance.md	Stock and currency tools
docs/howtos/communications.md	Gmail and messaging channels
docs/howtos/reverse.md	GitHub repo reverse-engineering
README.md	Project overview and quick start
SPEC.md	Full configuration schema reference
QUICKSTART.md	Quick start guide